

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Consistent 4-space indentation in app.py; uniform formatting in HTML/CSS templates.
2	Proper File Structure	Files and folders are logically organized	Partial	Dataset, Flask app, and Training notebook are separated, but model files sit inside the Flask folder rather than a dedicated models directory.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like applicant_income, credit_history, loan_amount clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Partial	Route names (home, predict, submit) are simple, but submit() handles preprocessing, scaling and prediction together.
5	Code Comments	Inline and block comments explain logic	Partial	app.py has section comments; the training notebook has comment-only headers with no detailed explanations.
6	Modular Design	Code is split into reusable functions/modules	No	Preprocessing, scaling and prediction logic are all written inline inside submit(); no reusable utility functions/classes.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Training notebook repeats the model-saving and scaler-saving cells twice.
8	Error Handling	Exceptions and errors are handled gracefully	No	No try/except blocks; invalid or missing form input will raise an unhandled exception.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Random Forest Model (rdf.pkl)	Python / scikit-learn	Loaded in app.py to predict loan eligibility in the /submit route	Medium

2	StandardScaler (scale1.pkl)	Python / scikit-learn	Scales applicant input features before prediction in the /submit route	Medium
3	style.css	CSS	Shared styling across home.html, predict.html and submit.html	High
4	app.py Flask routes	Python / Flask	Defines the home / predict / submit request-handling pattern	Low
5	Data preprocessing steps	Python / pandas	Encoding & cleaning logic in the training notebook (not yet extracted into a reusable script)	Low

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clean, consistent formatting across app.py and templates.
Readability	4	Descriptive variable names; minimal but useful comments.
Reusability	2	Logic is not modularized into reusable functions or classes.
Documentation / Comments	3	Some inline comments; lacks docstrings and notebook markdown explanations.
Overall Score	3	Solid foundational structure; would benefit from modular functions and error handling.